

A Bidirectional Neuro-Symbolic Foundation Pipeline for Continuous Sign Language AI

Closed-Loop Spatial Scaffolding, Causal Memory, and Anatomy-Aware Diffusion

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Executive Abstract & Project Vision

Continuous Sign Language Processing—spanning forward Sign Language Production (SLP: Text-to-Sign) and backward Sign Language Translation (SLT: Sign-to-Text)—represents a fundamental frontier in multimodal artificial intelligence and biomechanical digital twin modeling. Current computational paradigms exhibit severe structural limitations. In the forward pass, systems oscillate between rigid procedural avatars that lack natural biological momentum and unconstrained deep generative models (e.g., Pose-VQVAE, discrete diffusion) that suffer from quantization blur, spatial drift, and the smearing of centimeter-scale phonological minimal pairs. In the backward pass, streaming neural translators suffer from **Spatial State Collapse**—wherein sliding temporal context windows purge historical 3D spatial reference loci (R -loci), breaking directional verb agreement—and **Environmental Grounding Blindness**, failing to resolve polysemic depicting classifiers (such as handshake CL: C) without visual scene grounding. Crucially, current literature treats production and translation as disconnected unidirectional tasks.

This doctoral project establishes a unified, closed-loop **Bidirectional Neuro-Symbolic Foundation Model** for continuous sign language AI. In the **Forward Pass (Text-to-Sign)**, we decouple symbolic syntax from continuous physical naturalization across four specialized stages: (1) an SVO-to-Time-Topic-Comment (TTC) transfer grammar maintaining a dynamic 3D Spatial Registry with generative LLM procedural fallback for out-of-vocabulary terms; (2) a modular sub-lexical procedural engine compiling phonological primitives into a deterministic 52-joint SMPL-X skeletal scaffold ($\hat{\mathbf{P}}$) via closed-form 2-bone arm Inverse Kinematics (IK), 15-DOF 3-phalanx hand anatomy, Kapandji limits, and a quartic Biomechanical Strain/Pain Objective (P_{strain}); (3) a Continuous Residual Diffusion Transformer ($\mathbf{X}_{\text{final}} = \Pi_{\mathcal{M}}(\hat{\mathbf{P}} + \Delta\mathbf{X}_{\theta})$) projected onto the kinematic manifold to guarantee bone-length invariance and protect millimetric contact bounds ($r_{\text{tol}} \leq 5$ mm); and (4) Adaptive Layer Normalization (AdaLN) for persona prosody and cross-lingual transfer. In the **Backward Pass (Sign-to-Text)**, we resolve streaming latency and environmental blindness via: (1) an anatomical perceptual front-end recovering 15-DOF hands and FLAME blendshapes in an invariant sternum frame; (2) continuous 3D loci raycasting from palm aiming and ocular gaze into an external, stateful **Causal Spatial Memory Bank** (M_t) decoupled in $\mathcal{O}(1)$ space from the LLM context window with deictic parity inversion ($\mathbf{P}_{\text{deictic}}$); and (3) an autoregressive decoder integrating *SceneGenie* 3D semantic scene graphs for real-world object grounding. Both modalities are unified by the **Spatial-SignLLM**, an autoregressive foundation model defined over a joint vocabulary $\mathcal{V}_{\text{total}} = \mathcal{V}_{\text{text}} \cup \mathcal{V}_{\text{phono}} \cup \mathcal{V}_{\text{spatial}}$ optimized via standard causal language modeling. We outline a phased 4-year plan under Aalto's **3-Paper Cumulative Strategy** (CVPR $\rightarrow$ ACL $\rightarrow$ NeurIPS) leveraging CSC's **EuroHPC LUMI supercomputer** and participatory co-design with the Finnish Association of the Deaf (*Kuurojen Liitto ry*).

1 Introduction, Societal Motivation & Global Human Impact

Over **70 million Deaf and hard-of-hearing individuals** worldwide communicate primarily through natural sign languages [11, 39]. Despite transformative advances in natural language processing (NLP) and large multimodal models (LMMs) for spoken communication, automated sign language technologies remain severely underdeveloped. Deaf citizens face persistent barriers in healthcare consultations, civic institutions, higher education, and digital services. Within the European Union, this gap conflicts with the European Accessibility Act (Directive 2019/882) and national legislation—most notably Finland's Sign Language Act (*Viittomakielilaki 352/2015*), which codifies the statutory right of signers to access public services in their mother tongue.

The core computational challenge of sign language stems from its physical medium. Sign language is not a 1D acoustic waveform or a sequence of linear text tokens; it is a **multi-channel**,

asynchronous, 4D spatio-temporal broadcast operating across the 3D volume surrounding the human body over continuous time [11, 24, 39]. An authentic signing interaction coordinates four simultaneous visual channels:

1. **Manual Articulations:** Rapid finger configurations (15 rotational degrees of freedom per hand), palm orientations, and continuous wrist trajectories in 3D Euclidean space.
2. **Non-Manual Signals (NMS):** Sub-pixel facial Action Units (eyebrow elevation AU1+AU2, furrowing AU4, mouth morphemes, ocular gaze vectors) that convey grammatical prosody, negation, and clause boundaries [31, 32].
3. **Torso and Head Posture:** Spatial body shifts and head tilts marking discourse perspective and role shifting [26].
4. **3D Referential Space (*R-loci*):** The programmatic projection of invisible 3D coordinate anchors into the air around the torso to represent discourse entities, which directional agreement verbs physically interpolate between to govern grammatical agency [27, 33].

Automating continuous sign communication requires solving two coupled problems: **Forward Generative Production (SLP: Text-to-Sign)**, synthesizing continuous, biologically natural, and grammatically sound 3D anatomical skeletal streams from text; and **Backward Perceptual Translation (SLT: Sign-to-Text)**, ingesting continuous video, tracking 3D spatial discourse anchors under streaming latency constraints (< 300 ms), and decoding fluent text while grounding gestures to the physical environment.

Historically, these tasks have been treated in isolation with monolithic deep neural networks. Because high-fidelity 3D motion-capture sign datasets are scarce (~ 80 hours globally), generative models suffer from anatomical hallucinations and quantization blur that smear phonological minimal pairs [43]. Conversely, streaming translators suffer from **Spatial State Collapse**, wherein sliding temporal windows purge the spatial memory required to resolve directional pronouns [23, 33]. Hosted at the **ELLIS Institute Finland** and **Aalto University** under the supervision of **Dr. Azade Farshad**, this 4-year doctoral project establishes a bidirectional neuro-symbolic pipeline that guarantees structural grammatical integrity through deterministic procedural scaffolding while mastering physical fluidity, causal spatial memory, and foundation modeling through continuous diffusion and the Spatial-SignLLM.

2 Linguistic Taxonomy & 3D Spatial Phonology

To satisfy the strict grammatical standards of native Deaf signers, computational representations must be grounded in formal sign language linguistics.

2.1 Sub-Lexical Parameter Composition & Minimal Pairs

In formal sign linguistics (Stokoe, 1960 [39]; Battison, 1978 [4]), every lexical sign is decomposed into five discrete phonological parameters:

$$S = \langle \mathbf{H}, \mathbf{L}, \mathbf{M}, \mathbf{O}, \mathbf{F}_{\text{NMS}} \rangle \quad (1)$$

where **H** represents Handshape (finger joint angles), **L** denotes 3D anatomical Location target, **M** is Movement trajectory primitive, **O** $\in \text{SO}(3)$ is wrist Orientation, and $\mathbf{F}_{\text{NMS}} = \langle \mathbf{b}, \mathbf{m}, \mathbf{g}, \mathbf{h} \rangle$ encodes facial Action Units (brows **b**, mouth **m**, gaze **g**, head tilt **h**).

Signs differing in exactly one parameter are **phonological minimal pairs**, where centimeter-scale deviations invert semantic meaning:

- **Location Minimal Pairs:** In American Sign Language (ASL), **MOTHER** is articulated with an open-5 handshape contacting the chin, whereas **FATHER** contacts the forehead. A vertical deviation of mere centimeters inverts maternal to paternal kinship.
- **Movement Minimal Pairs:** **FATHER** involves a single contact tap at the forehead, whereas **GRANDFATHER** shares the identical handshape and location but executes a double-arc outward ballistic bounce, requiring high-frequency temporal acceleration tracking.
- **Orientation Minimal Pairs:** **NAME** involves two H-handshapes tapping transversely; altering palm orientation from inward to downward produces an ungrammatical form.

2.2 Non-Manual Signals & 3D Spatial Syntax (R-loci)

Non-manual signals act as the visual equivalent of spoken prosody and syntactic clause delimiters [31,32]: raised eyebrows (AU1+AU2) mark Yes/No questions and conditionals, furrowed eyebrows (AU4) mark Wh-questions, and ocular gaze (g_{gaze}) raycasts toward empty spatial loci to establish pronominal reference before manual indexing [11,38].

As modeled by Pfau et al. (2018) [33] and Lillo-Martin & Meier (2011) [27], signers project coordinate anchors into the air around their body, termed **Referential Loci (R-loci)**. In our interaction volume:

- ▶ Locus₁ (Signer Sternum): [0.00, 0.00, 0.00] m, designating first-person (“I”, “Me”).
- ▶ Locus₂ (Addressee): [0.00, 0.60, 0.00] m, designating second-person (“You”).
- ▶ Locus_{3a}, Locus_{3b} (Third-Party Referents): Right hemisphere [+0.35, 0.45, +0.05] m and Left hemisphere [−0.35, 0.45, +0.05] m.

Directional verbs physically traverse these loci: **Regular verbs** (GIVE, VISIT) move from Subject to Object (Locus_{3a} → Locus₁ for “Mother visits me”); **Backward verbs** (INVITE, TAKE) invert physical trajectory relative to agency, moving from Goal to Agent (Locus_{3a} → Locus₁ for “I invite Mother”) [29,33]; and **Orientation-only verbs** (LOOK-AT, WARN) aim palm or index vectors directly toward the active referent.

2.3 Empirical Regularities in Sign Lexicons (ASL-LEX 2.0)

Mentored by Dr. Marco Marini (Sapienza University of Rome) to formalize sub-lexical decomposition, our parameter space is calibrated against **ASL-LEX 2.0** (2,723 signs across 54 dimensions [6,36]). Prior linguistic analyses reveal critical structural constraints:

1. **Handshape Distribution Skew:** In accordance with Battison’s phonological laws [4], the top 6 unmarked handshapes (B, A/S, 1, 5, 0, C) represent $\approx 52\%$ of the lexicon, and the top 12 cover $\approx 78\%$. Calibrating 12 anatomical finger configurations provides immediate coverage of conversational vocabulary.
2. **Spatial Location Clustering:** Signs cluster tightly: $\approx 40\%$ occur in neutral space before the chest, $\approx 34\%$ cluster around cranial/facial landmarks (chin, forehead, cheek), and the remainder occur on the trunk or base hand, motivating millimetric precision on cranial zones.
3. **Battison’s Symmetry Constraints:** Two-handed asymmetric signs restrict the passive base hand to unmarked resting shapes, dramatically simplifying multi-hand coordination.
4. **Movement Dynamics:** Approximately 62% of signs involve single continuous paths (verbs), while 38% feature repeated contact taps or oscillations (nouns), parameterized via a repetition multiplier.

3 Literature Review & Rigorous Scientific Gap Analysis

3.1 Generative Production (SLP): Procedural Scaffolding vs. Neural Diffusion

Generative SLP has oscillated between procedural correctness and perceptual realism. Early procedural animation systems (VisiCast, JASigning) converted phonetic scripts (HamNoSys/SiGML) into deterministic avatars [28]. While guaranteeing bone-length invariance and joint limits, they produced robotic motion lacking biological momentum and co-articulation. Recent neural approaches using autoregressive Transformers, GANs, or discrete latent diffusion (e.g., G2P-DDM [43], Neural Sign Actors [3]) mapped glosses to vector-quantized latent spaces (Pose-VQVAE). However, codebook quantization introduces spatial blur that smears millimeter-scale minimal pairs. Conversely, continuous unconstrained skeletal diffusion models (e.g., MDM [41]) diffuse in Euclidean space $\mathbb{R}^{J \times 3}$; lacking kinematic manifold projection, they stretch bone lengths (“rubber fingers”) and breach physiological Range-of-Motion (ROM) limits. Simulator-guided diffusion (e.g., Phys-Diff [45], BioMoDiffuse) integrates rigid-body physics (Isaac Gym) for gross locomotion, but non-differentiable rigid simulators cannot resolve high-frequency 15-DOF finger dexterity, Kapandji thumb opposition, or sub-centimeter ($r_{\text{tol}} \leq 5 \text{ mm}$) cranial contact boundaries. Similarly, while recent transition diffusion architectures like StgcDiff [21] apply spatial-temporal graph

convolutions as soft neural conditioning, soft priors cannot mathematically guarantee bone-length conservation or prevent anatomical dislocations. Appearance-level models (e.g., DiffSign [25]) render photorealistic 2D RGB pixels from 2D poses, bypassing 3D kinematic fidelity. Our solution is **Manifold-Projected Continuous Residual Diffusion** ($\mathbf{X} = \Pi_{\mathcal{M}}(\hat{\mathbf{P}} + \Delta\mathbf{X}_{\theta})$), anchoring diffusion to exact procedural coordinates while enforcing iterative holonomic manifold constraints.

3.2 Perceptual Translation (SLT) & Multimodal Sign Foundation Models

Current SOTA Sign Language Translation treats the mapping from video to text as an end-to-end sequence-to-sequence problem ($\mathbf{V}_{1:T} \rightarrow \mathbf{W}$). In conversational settings, existing models exhibit three fundamental structural failure modes:

1. **Spatial State Collapse (The Streaming Bottleneck):** As independently diagnosed by Kermani et al. [23] and Ranum et al. [34], real-time streaming translation (< 300 ms latency) requires sliding temporal attention windows [44]. Discarding historical frames purges established R -loci, breaking long-range coreference and causing directional agreement verbs to hallucinate incorrect semantic arguments [33]. While recent prompting heuristics (e.g., SpaMo [37]) append frame-level 2D/3D spatial distances to LLM prompts, static text prompting lacks a stateful, persistent 3D spatial register that persists across multi-turn conversational dialogue.
2. **Environmental Grounding Blindness:** Pure-skeletal translators [10, 44] are blind to external physical space. When a signer points to a room object or uses an ambiguous depicting classifier (e.g., $\text{CL}:\text{C}$, representing a cup, cylinder, or vase) [40], skeletons lack visual grounding to identify the referent.
3. **Unidirectional Disconnection in Foundation Sign LLMs:** Recent multimodal foundation efforts (e.g., SignLLM [12], ViPo-MLLM, SignBind-LLM) explore multilingual text-to-pose generation or visual-pose alignment. However, models such as SignLLM generate 2D/3D skeletal keypoints directly from text prompts without 3D spatial referential loci, causing token explosion ($> 3,000$ values/s) and phonetic minimal-pair smearing, while remaining strictly unidirectional.

Research Challenge	Current SOTA Paradigm	Structural Limitation
Quantization Blur	Pose-VQVAE + Discrete Diffusion (G2P-DDM [43])	Codebook quantization rounds continuous coordinates, smearing minimal pairs.
Unconstrained Drift	Unconstrained Motion Diffusion (MDM [41])	Euclidean diffusion without kinematic manifold stretches bones (“rubber fingers”).
Locomotion vs. Dexterity	Simulator-Guided Diffusion (PhysDiff [45])	Rigid-body simulators model gross locomotion, failing on 15-DOF dexterity and cranial contacts.
Soft Graph Priors	Graph-Conditioned Diffusion (StgcDiff [21])	Soft ST-GCN conditioning cannot mathematically guarantee bone-length invariance (ℓ_{ij}).
Spatial State Collapse	Sliding-Window / Heuristics (Kermani [23], SpaMo [37])	Truncating context windows or static prompting purges historical R -loci across turns.
Environmental Blindness	Pure-Skeletal Translators (STMC [44])	Skeletons lack environmental context to resolve depicting classifiers ($\text{CL}:\text{C}$).
Unidirectional Generation	Text-to-Pose Foundation Models (SignLLM [12])	Directly outputs raw joints; lacks 3D loci memory (M_t), kinematics, and bidirectional closed-loop.

Table 1: Core Algorithmic Challenges and SOTA Failure Modes in Sign Language AI.

4 Proposed Technical Methodology: The Closed-Loop Pipeline

4.1 Architectural Blueprint & Functional Decoupling

The fundamental insight of our architecture is **functional decoupling**:

- **Symbolic & Kinematic Algorithms** guarantee hard physical laws, anatomical joint limits, collision avoidance, and grammatical spatial indexing via **deterministic procedural compilers and closed-form analytical IK**.
- **Continuous Generative Neural Models** synthesize biological fluidity, micro-coarticulation, stylistic prosody, and commonsense reasoning via **continuous residual diffusion models and the Spatial-SignLLM**.

As shown in Figure 1, the pipeline couples the Forward Production Pass (SLP), the Backward Perceptual Pass (SLT), and the Spatial-SignLLM Multimodal Bridge into a closed-loop system.

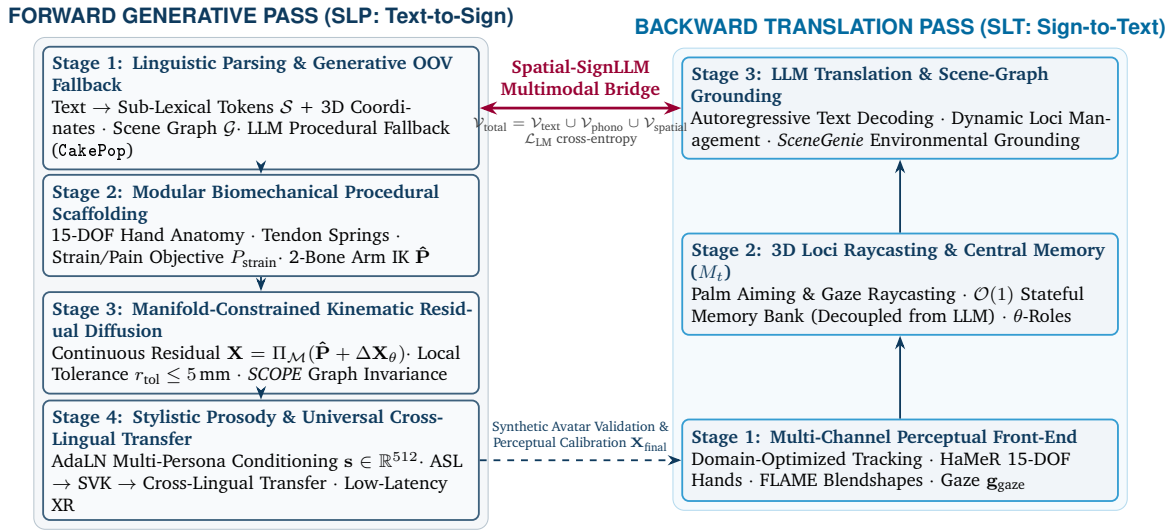


Figure 1: The Unified Bidirectional Spatial Foundation Pipeline with the Spatial-SignLLM Multimodal Bridge.

4.2 The Forward Pass: Text-to-Sign Language Production (SLP)

The Forward Pass transforms spoken text into a continuous 60 Hz 3D avatar stream across four specialized stages:

4.2.1 Stage 1: Linguistic Parsing, 3D Spatial Registry & Generative LLM OOV Fallback

Natural sign languages function as Topic-Prominent languages governed by **Time-Topic-Comment (TTC)** structure [17, 26, 31]. Feeding linear spoken Subject-Verb-Object (SVO) text directly to an avatar produces ungrammatical pidgin. Stage 1 executes three operations:

1. **Syntactic Transfer Grammar (SVO $\rightarrow$ TTC):** Extracts temporal adverbials and fronts them to clause-initial position to establish the discourse timeline, deletes non-signing copulas (*is*, *are*, *was*), and shifts interrogative markers clause-final where facial markers (AU4) frame the proposition.
2. **Coreference Resolution & 3D Spatial Registry:** Allocates coordinate anchors in the signer's interaction volume: $\text{Locus}_1 = [0.00, 0.00, 0.00] \text{ m}$ (Self), $\text{Locus}_{3a} = [+0.35, 0.45, +0.05] \text{ m}$ (Primary referent), and $\text{Locus}_{3b} = [-0.35, 0.45, +0.05] \text{ m}$ (Secondary referent). Directional verbs bind trajectory endpoints directly between these coordinates. The sentence is compiled into a 3D spatial scene graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ encoded via a Relational Graph Transformer (RGT).
3. **Generative LLM Out-of-Vocabulary (OOV) Procedural Fallback:** Rather than freezing or falling back to tedious 8-second letter fingerspelling when encountering unseen terms [8, 30], we leverage the spatial commonsense reasoning of LLMs to synthesize **Depicting Signs (Classifier Predicates)** [11, 40] from atomic sub-lexical parameters. Crucially, the LLM does not emit

continuous joint Euler angles directly; rather, it outputs high-level discrete phonological tokens—handshape primitives (**H**), movement paths (**M**), palm orientations (**O**), and contact locations (**L**)—which are subsequently validated against phonological collision rules and compiled into continuous kinematics by Stage 2.

Listing 1: Validated Pydantic 4D Phonetic JSON Interface with Procedural OOV Synthesis.

```
{
  "sentence_id": "AALTO_SVK_001", "asl_gloss": "YESTERDAY MOTHER 3a-VISIT-1",
  "sentence_nms": {"eyebrows": "NEUTRAL", "negation": false},
  "spatial_registry": {
    "Locus_3a": {"entity": "MOTHER", "coord": [0.35, 0.45, 0.05]},
    "Locus_1": {"entity": "SELF", "coord": [0.00, 0.00, 0.00]}
  },
  "tokens": [
    {"lemma": "YESTERDAY", "handshape": "HS_A_THUMB", "loc": [0.15, 0.10, 0.30], "mov": "CHEEK_TAP"},
    {"lemma": "MOTHER", "handshape": "HS_5_OPEN", "loc": [0.00, 0.08, 0.20], "mov": "CHIN_TAP"},
    {"lemma": "VISIT", "handshape": "HS_V_ALT", "start": [0.35, 0.45, 0.05], "end": [0.00, 0.00, 0.00]},
    {"lemma": "CAKEPOP", "oov_fallback": true, "procedural_tokens": {
      "idx": "CURVED", "mid": "CLOSED", "rng": "CLOSED", "pnk": "EXTENDED_SPREAD_R", "thm": "THM_PINCH_INDEX"}
    }
  ]
}
```

4.2.2 Stage 2: Modular Sub-Lexical Procedural Scaffolding & Biomechanical Muscle Engine

Stage 2 compiles the 4D JSON stream into a deterministic 52-joint SMPL-X skeletal scaffold $\hat{\mathbf{P}} \in \mathbb{R}^{T \times 52 \times 3}$ via analytical closed-form IK in $\mathcal{O}(1)$ time, guaranteeing anatomical feasibility before neural processing:

- 15-DOF 3-Phalanx Hand Model & Musculature:** Fingers are modeled as authentic 3-phalanx kinematic chains: Metacarpophalangeal (MCP: 2-DOF flex $[-15^\circ, 90^\circ]$, abd $[-20^\circ, 20^\circ]$), Proximal Interphalangeal (PIP: 1-DOF flex $[0^\circ, 100^\circ]$), and Distal Interphalangeal (DIP: 1-DOF flex $[0^\circ, 80^\circ]$). The thumb articulates across Trapeziometacarpal/CMC (2-DOF), MCP (1-DOF), and Interphalangeal/IP (1-DOF) joints, with modeled thenar and hypothenar muscular pads.
- Quartic Articular Range-of-Motion (ROM) Barrier Objective $P_{\text{strain}}(\theta)$:** Human motor control minimizes joint strain. We formalize this biological principle via an explicit quartic pain/strain penalty:

$$P_{\text{strain}}(\theta) = \sum_{j=1}^J \left(\frac{\theta_j - \theta_{j,\text{rest}}}{\theta_{j,\text{max}} - \theta_{j,\text{min}}} \right)^4 \quad (2)$$

Escalating sharply near physiological limits, resolving motion along the gradient of minimal strain naturally chooses the path of least physical resistance, preventing awkward joint hyper-extensions. While the skeletal scaffold $\hat{\mathbf{P}}$ is initialized via analytical closed-form IK, $P_{\text{strain}}(\theta)$ serves as a smooth non-linear interior barrier function defining the allowable kinematic manifold $\mathcal{M}$, evaluated during iterative manifold projection $\Pi_{\mathcal{M}}$ and offline trajectory optimization. Dynamic trajectory smoothness and minimum-jerk optimization ($\int \|\ddot{\mathbf{x}}\|^2 dt$) [18] are governed in continuous time by the residual diffusion score network.

- Multi-Digit Coordination & Soft-Body Collision Repulsion:** When distance between phalangeal segments falls below combined radii ($r_A + r_B$), an elastic repulsion vector prevents self-intersection. Dynamic priority scheduling orders digit flexion in folded signs (e.g., in M, N, T, pinky and ring curl before the thumb crosses and index folds over).
- Analytical Arm Kinematics with 2-DOF Clavicle & Humeral Swivel Orbit:** Given shoulder root $\mathbf{x}_{\text{shoulder}}$ and wrist $\mathbf{x}_{\text{wrist}}(t)$, the shoulder anchor is first modulated by a 2-DOF clavicle articulation model (15° – 20° elevation and protraction) to prevent reach singularities at cranial minimal-pair targets (e.g., FATHER at the forehead). Let $\vec{d} = \mathbf{x}_{\text{wrist}} - \mathbf{x}_{\text{clavicle}}$ ($d = \|\vec{d}\|$), with upper arm length L_1 and forearm length L_2 . The elbow flex angle θ_{elbow} is resolved in closed

form via the Law of Cosines:

$$\theta_{\text{elbow}} = \arccos \left(\text{clamp} \left(\frac{d^2 - L_1^2 - L_2^2}{2L_1L_2}, -1, 1 \right) \right) \quad (3)$$

Humeral swivel redundancy $\phi \in [0, 2\pi)$ parameterizes elbow position along an orbit circle orthogonal to $\vec{u} = \vec{d}/d$, referenced to a biological resting plane with singularity damping at full arm extension.

5. **Facial Blendshape Binding (F_{NMS}):** Non-manual clause modifiers are mapped into 50 FLAME blendshapes synchronized with the manual stroke phase.

4.2.3 Stage 3: Manifold-Constrained Kinematic Residual Diffusion

Pure procedural animation is structurally correct but mechanically stiff. Stage 3 employs continuous diffusion solely to synthesize biological momentum, micro-coarticulation, and natural fluidity atop the rigid scaffold:

$$\mathbf{X}_{\text{final}} = \Pi_{\mathcal{M}}(\hat{\mathbf{P}} + \Delta\mathbf{X}_{\theta}) \quad (4)$$

where $\Delta\mathbf{X}_{\theta}$ is the predicted residual micro-displacement. Grounded in **Dr. Farshad's methodologies in topologically constrained generative models** (*SCOPE* [15], *SceneGenie* [13]), we enforce Riemannian manifold projection ($\Pi_{\mathcal{M}}$) via iterative holonomic constraint projection using the SHAKE/RATTLE molecular dynamics algorithm [2, 35] (converging in ≤ 3 –5 iterations) adapted to skeletal kinematic trees:

1. **Bone-Length Invariance ($\mathcal{L}_{\text{graph}}$):** Enforces exact distance constraints ($\|\mathbf{x}_i(t) - \mathbf{x}_j(t)\|_2 = \ell_{ij}$) across all 52 skeletal connections and 15 finger phalanges, permanently eliminating bone stretching.
2. **Character-Adaptive Contact Tolerance (r_{tol}):** Enforces tight millimetric bounds ($r_{\text{tol}} \leq 5$ mm) at critical contact landmarks (e.g., chin vs. forehead for MOTHER vs. FATHER), while permitting natural compliance (20–30 mm) during unconstrained movement in neutral space.
3. **Pain Envelope Preservation:** Ensures neural residual updates respect the joint limit and strain boundaries established by $P_{\text{strain}}(\theta)$.

The Diffusion Transformer backbone is pretrained on **Motion-X++** (19.5M frames) [46] for general physical dynamics, followed by LoRA fine-tuning on How2Sign [10].

4.2.4 Stage 4: Stylistic Prosody & Universal Cross-Lingual Transfer

A style vector $\mathbf{s} \in \mathbb{R}^{512}$ modulates the denoiser via Adaptive Layer Normalization (AdaLN) to synthesize distinct signer personas, signing tempos, and emotional prosody. Swapping sub-lexical dictionary tables enables zero-shot cross-lingual transfer from ASL to Finnish Sign Language (SVK) and other sign languages with a detailed enough sub-lexical dictionary.

4.3 The Backward Pass: Sign-to-Text Language Translation (SLT)

The Backward Pass perceives and translates continuous signing video back into spoken text through three interpretable stages:

4.3.1 Stage 1: Multi-Channel Perceptual Front-End & Domain-Optimized Tracking

Operating over dedicated signers facing the camera, the perceptual front-end recovers 3D anatomical articulators:

1. **15-DOF Hand Mesh Reconstruction:** High-dexterity tracking via HaMeR [47] extracts local rotational joint angles across all 15 phalangeal degrees of freedom:

$$\mathbf{q}_{\text{hand}} = \{\theta_{\text{MCP}}, \theta_{\text{PIP}}, \theta_{\text{DIP}}, \phi_{\text{abduction}}\}_{i=1}^5 \in \mathbb{R}^{15} \quad (5)$$

2. **Facial Action Units & Gaze Vector:** Tracks 50 FLAME blendshapes (eyebrows, mouth morphemes) and the 3D ocular gaze unit vector $\mathbf{g}_{\text{gaze}}(t) \in \mathbb{R}^3$.
3. **Body-Centric Sternum Normalization Frame:** Skeletal keypoints $\mathbf{x}_{\text{world}} \in \mathbb{R}^3$ are normalized to an invariant coordinate frame centered at the signer's sternum $([0, 0, 0])$ and scaled by biacromial shoulder width (W_{shoulder}), eliminating perspective and scale variances.

4.3.2 Stage 2: 3D Spatial Loci Raycasting & Central Memory Bank (M_t)

Stage 2 tracks spatial interactions and maintains discourse memory across turns:

- **Palm & Gaze Aiming Raycasting:** Continuous 3D aiming rays are cast from the normalized palm orientation vector ($\vec{v}_{\text{aim}} = \mathbf{R}_{\text{wrist}} \cdot \mathbf{u}_{\text{point}}$) and ocular gaze vector $\mathbf{g}_{\text{gaze}}(t)$, intersecting active 3D spatial loci in the interaction volume.
- **Decoupled Central Memory Bank (M_t):** Active referents ($K \leq 8$ entities) are stored in an external stateful memory bank:

$$M_t = \{\mathbf{m}_k = \langle \mathbf{x}_k, \mathbf{e}_k, \alpha_k, \tau_k \rangle\}_{k=1}^K \quad (6)$$

where $\mathbf{x}_k \in \mathbb{R}^3$ is the physical coordinate anchor, $\mathbf{e}_k$ is the entity embedding, α_k is the salience weight, and τ_k is the last interaction timestamp. Decoupling M_t into an $\mathcal{O}(1)$ external register allows entities to persist across long dialogues without inflating the LLM context or causing spatial state collapse when sliding temporal windows advance. Crucially, while articulatory tracking is normalized to the sternum for scale invariance, spatial referents $\mathbf{x}_k$ in M_t are parameterized in an egocentric horizontal ground-plane frame decoupled from torso yaw, preventing discourse anchors from erroneously rotating when a signer executes a physical role-shift (15° – 30° torso turn).

- **Deictic Frame-of-Reference Inversion ($\mathbf{P}_{\text{deictic}}$):** In face-to-face dialogue, the coordinate frames of signer and avatar are rotated by 180° around the vertical axis. To prevent semantic left-right inversion during turn-taking, we apply an explicit Deictic Perspective Parity Operator:

$$\mathbf{x}_{\text{avatar}} = \mathbf{P}_{\text{deictic}} \cdot \mathbf{x}_{\text{signer}}, \quad \mathbf{P}_{\text{deictic}} = \text{diag}(-1, 1, 1) \quad (7)$$

- **Spatial Hysteresis & Thematic θ -Role Matching:** To mitigate high-frequency monocular tracking noise without sacrificing precision, aiming rays are temporally integrated across a 5–10 frame window (~ 160 ms). Instead of a naive rigid threshold, intersection with active referents in M_t is resolved via an adaptive probabilistic Gaussian tolerance kernel $\mathcal{N}(\mathbf{x}_k, \Sigma_{k,t})$, where spatial covariance $\Sigma_{k,t}$ is updated dynamically by an Extended Kalman Filter (EKF). Furthermore, phonological contrast in natural sign discourse ensures that distinct entities inhabit well-separated canonical signing sectors (contralateral, center, ipsilateral) with angular separation $> 30^\circ$, reliably exceeding the monocular tracking uncertainty cone. Directional verbs are resolved across three phonological classes [29]: Path verbs (GIVE) via trajectory start/end ($\mathbf{p}_{\text{start}} \rightarrow \mathbf{p}_{\text{end}}$); Backward verbs (INVITE) via lexical mapping inverting Goal to Source; and Orientation verbs (LOOK-AT) via $\vec{v}_{\text{aim}}$.

4.3.3 Stage 3: LLM Translation, Memory Management & Scene-Graph Grounding

Stage 3 decodes fluent text while governing memory and grounding gestures in the physical environment:

- **Active Memory Governance:** The LLM manages M_t dynamically: registering new loci $\mathbf{x}_{\text{new}}$ upon introduction, refreshing salience α_k upon re-reference, and evicting completed topics via soft exponential decay $\alpha_k(t)$ in $\mathcal{O}(1)$ time.
- **Environmental Grounding via Semantic Scene Graphs (*SceneGenie* [13]):** When a signer points to a real-world object (e.g., a coffee cup, door, or medical tool) or uses an ambiguous polysemic depicting classifier ($\text{CL} : \mathcal{C}$), skeletal tracking cannot disambiguate the referent. We structure the room environment into a 3D semantic scene graph $\mathcal{G}_{\text{env}} = (\mathcal{V}_{\text{obj}}, \mathcal{E}_{\text{rel}})$. Intersecting the pointing ray $\vec{v}_{\text{aim}}$ with object bounding volumes allows the LLM to instantly resolve real-world referents without heavy raw video compute.

4.4 The Spatial-SignLLM: Unified 3D Spatial-Lexical Foundation Modeling

Bidirectional synergy between continuous physical kinematics and symbolic linguistics is unified through an autoregressive **Spatial Sign Language Large Model (Spatial-SignLLM)**. Rather than attempting to train high-variance reinforcement learning policies or unstable GAN discriminators across the continuous 52-joint boundary, we treat spatial sign communication as a **hierarchical multimodal language modeling task**.

Discrete 3D Coordinate Tokenization: Drawing on embodied robotics and vision-language-action models (RT-2 [5], PaLM-E [9], SpatialVLM [7]), large Transformer backbones natively ingest and emit physical spatial coordinates when continuous 3D space is discretized into coordinate tokens. While recent architectures such as SignLLM [12] pioneered multilingual text-to-pose generation via LLMs, they autoregressively emit raw joint coordinates directly without 3D spatial referential memory, leading to severe token explosion ($> 3,000$ values/s) and minimal-pair blurring. In contrast, our **Spatial-SignLLM** unifies spatial linguistics and biomechanics by defining a multimodal spatial-lexical vocabulary:

$$\mathcal{V}_{\text{total}} = \mathcal{V}_{\text{text}} \cup \mathcal{V}_{\text{phono}} \cup \mathcal{V}_{\text{spatial}} \quad (8)$$

where $\mathcal{V}_{\text{text}}$ is the standard subword vocabulary for spoken text, $\mathcal{V}_{\text{phono}}$ is the sub-lexical parameter alphabet $\mathcal{S}$ representing discrete handshapes and movement primitives, and $\mathcal{V}_{\text{spatial}}$ denotes discrete coordinate tokens partitioning the canonical interaction volume $[-1, 1]^3$ into spatial bins:

$$\mathbf{z}_{\text{locus}} = \text{Embed}_{\text{spatial}}(\langle x, y, z \rangle) \in \mathbb{R}^{d_{\text{model}}} \quad (9)$$

The entire foundation model is trained end-to-end via standard causal autoregressive language modeling cross-entropy over discrete token sequences:

$$\mathcal{L}_{\text{LM}} = - \sum_{i=1}^N \log P(w_i \mid w_{<i}), \quad w_i \in \mathcal{V}_{\text{total}} \quad (10)$$

How It Operates in the Bidirectional Loop:

1. **Forward Direction (Macro-Compiler at 1–2 Hz):** Direct autoregressive prediction of 60 Hz joint torques causes catastrophic token explosion ($60 \text{ frames/s} \times 52 \text{ joints} = 3,120 \text{ values/s}$). The Spatial-SignLLM resolves this by operating exclusively at the **discourse rate (1–2 Hz)**, emitting structured token sequences per concept:

[TIME: YESTERDAY] $\rightarrow$ [LOC_3a: MOTHER, (+0.35, 0.45, +0.05)] $\rightarrow$ [VERB: VISIT, 3a $\rightarrow$ 1]

When encountering unseen terms (e.g., 'CakePop'), the model emits discrete procedural depicting tokens (PINCH_INDEX, pinky spread) rather than continuous joint angles. This discrete 1–2 Hz blueprint parameterizes Stage 2 (closed-form 15-DOF procedural IK) and Stage 3 (residual diffusion), which synthesize smooth 60 Hz physics. The LLM acts as the “planner” while the kinematic pipeline acts as the “motor execution system.”

2. **Backward Direction (Sign-to-Text Decoding):** In reverse, our video perceptual front-end and memory bank M_t extract 15-DOF handshapes S_t and tracked 3D coordinate tokens $\mathcal{V}_{\text{spatial}}$ from video. The Spatial-SignLLM ingests these multimodal tokens; self-attention heads bind the spatial trajectory endpoints back to the entities in M_t , correctly translating directional agreement verbs (VISIT, INVITE) into fluent spoken sentences without context window collapse.
3. **Optimization Stability on EuroHPC LUMI:** Pretraining across How2Sign and Motion-X++ (19.5M frames) with pseudo-grounded scaffolds extracted by our perception front-end uses standard cross-entropy, eliminating the high gradient variance of continuous policy gradients.

4.5 Empirical De-risking: Working 15-DOF Kinematic Prototype

To de-risk the technical methodology against concerns of doctoral tractability and demonstrate that the proposed neuro-symbolic pipeline is practically realizable, we have independently developed and deployed an interactive 15-DOF WebGL kinematics engine (anatomical_hand.html). Engineered deliberately as a lightweight, zero-dependency client-side prototype capable of executing smoothly across any consumer browser or mobile device, the engine compiles symbolic sub-lexical instructions into deterministic 3D hand configurations in closed-form $\mathcal{O}(1)$ time under anatomical joint limits and real-time tendon spring kinetics. While this interactive demonstrator proves topological feasibility on everyday hardware, the full non-convex quartic strain energy landscape

$P_{\text{strain}}(\theta)$ and multi-body capsule collision solvers are evaluated in offline batch optimization (PyTorch/Taichi) for large-scale training on EuroHPC LUMI.

Crucially, to validate the tractability of generative OOV depiction in practice, when prompted with the novel out-of-vocabulary concept 'CakePop' (a spherical confection atop a stick), the foundation LLM does not hallucinate raw continuous joint angles. Instead, it operates strictly at the discrete phonological level—reasoning over the object’s physical affordances to emit the discrete sub-lexical tokens:

$$\text{Tokens}(\text{CakePop}) = \{\text{idx} : \text{CURVED}, \text{mid} : \text{CLOSED}, \text{rng} : \text{CLOSED}, \\ \text{pnk} : \text{EXTENDED_SPREAD_R}, \text{thm} : \text{THM_PINCH_INDEX}\} \quad (11)$$

which are then deterministically compiled by the kinematic engine into anatomically valid 3D joint configurations, verified against formal depicting classifier grammar [40] and an ASL-LEX phonological filter to prevent illegal skeletal configurations. This working proof-of-concept establishes that the candidate already commands the foundational 3D graphics, analytical kinematics, and neuro-symbolic compilation infrastructure required to execute Track A and Track C without speculative startup delays.

5 Methodological Synergy with Target Lab at Aalto University & ELLIS Finland

Continuous sign language processing represents a high-impact application domain for **Dr. Azade Farshad’s** research group at **Aalto University / ELLIS Institute Finland**. As established in Section 4, our bidirectional neuro-symbolic framework directly integrates and extends her group’s pioneering methodologies in geometric deep learning and generative modeling:

1. **Scene-Graph Generative Conditioning (Extending *SceneGenie* [13])**: In *SceneGenie*, Dr. Farshad demonstrated that conditioning diffusion models on structured semantic scene graphs significantly improves visual coherence. Sign discourse syntax operates precisely as a 3D spatial scene graph $\mathcal{G} = (\mathcal{V}_{\text{entities}}, \mathcal{E}_{\text{relations}})$, where nodes encode coordinate loci (L_1, L_{3a}, L_{3b}) and edges encode directional agreement predicates. Encoded via a Relational Graph Transformer (RGT), these embeddings condition the residual diffusion denoiser, eliminating minimal-pair drift. In reverse, scene graphs ground physical room objects to disambiguate pointing gestures.
2. **Topologically Constrained Generative Modeling (Extending *SCOPE* [15] & *HieraSurg* [14])**: In surgical video generation and anatomical segmentation, Dr. Farshad established that biological structures require explicit graph-based continuity regularizers. High-DOF articulated hands and arms exhibit identical topological vulnerabilities during neural diffusion. Adapting these graph constraints guarantees bone-length invariance ($\mathcal{L}_{\text{graph}}$) and biological strain limits ($\mathcal{P}_{\text{pain}}$), preventing joint dislocations.
3. **ELLIS Priority “AI for Health & Society”**: According to the WHO, health encompasses complete physical, mental, and social well-being. Chronic communication isolation represents a severe social determinant of health. This bidirectional system acts as an **Assistive Communication Digital Twin**, dismantling systemic exclusion across healthcare, public services, and daily life.

6 Multi-Tier Evaluation Framework

To benchmark generative biomechanical accuracy (SLP) and translation fidelity (SLT), we establish a multi-tier framework spanning five complementary validation axes summarized in Table 2: (1) **Anatomical Landmark Precision** on millimeter-scale minimal pairs; (2) **Kinematic Motion Naturalism** via Fréchet Video Distance (FVD); (3) **Spatial Discourse Agreement** via Agreement Resolution Accuracy (ARA); (4) **Streaming Latency** under sub-100 ms real-time constraints; and (5) **Participatory Human Ecological Validity** evaluated with native Deaf signers from *Kuurojen Liitto ry*.

7 4-Year Doctoral Research Work Plan: Parallel Tracks & Staggered Milestones

Dimension	Evaluation Metric & Formulation	Design Target	Comparative Literature Baseline
Landmark Precision	Anchor-Fidelity Error (EFE_{mm}) : Euclidean error on fingertip/cranial key-points.	≤ 15 mm	G2P-DDM [43] (≈ 35 – 40 mm)
Spatial Grounding	Perplexity & Spatial Locus Alignment (SLA_{mm}) : Causal LM perplexity and 3D coordinate precision.	$\text{PPL} \leq 8.5$, $\text{SLA} \leq 12$ mm	Unconditioned LLMs / Disconnected tokenizers
Kinematic Realism	Fréchet Video Distance (FVD) and Kinematic Jerk ($\ \ddot{\mathbf{x}}\ ^2$).	$\text{FVD} \leq 40.0$	Human Motion Diffusion (MDM) [41]
Minimal-Pair Accuracy	Spatio-temporal classifier distinguishing minimal pairs (MOTHER vs. FATHER).	$\geq 92\%$ accuracy	Unconstrained generation baselines ($< 70\%$)
Agreement Resolution	Agreement Resolution Accuracy (ARA) : Subject-object assignment on directional verbs.	$\geq 92\%$ accuracy	Sliding-window SLT models [44] ($\approx 64\%$)
Continuous Translation	BLEU-4 & WER : Evaluated across continuous benchmarks (How2Sign, PHOENIX-2014T).	$\text{BLEU-4} \geq 28.5$, $\text{WER} \leq 24.0$	CorrNet+ [22], STMC [44]
Inference Latency	End-to-end streaming latency on modern GPU hardware.	< 100 ms (> 60 fps)	Dense Vision-Language Models (> 400 ms)
Everyday Chat MOS	Native Deaf User Evaluation: Likert scale (1–5) on Conversational Fluency and Spatial Clarity.	$\text{MOS} \geq 4.2/5.0$	Procedural animation avatars [28]

Table 2: Multi-Tier Evaluation Matrix balancing linguistic precision, foundation model perplexity, and physical realism.

Adhering to Aalto University’s articles-based doctoral dissertation model (requiring 3–4 top peer-reviewed publications), the research progresses across two **Concurrent Development Tracks (Track A: Kinematic Synthesis; Track B: Perceptual Tracking)** that execute in parallel from Month 1, coupled via the shared sub-lexical vocabulary $\mathcal{S}$ and body-centric frame, converging into the foundation model (**Track C**):

- **Years 1–2: Parallel Foundations (Kinematic Synthesis & Streaming Perception)**
 - *WP1 (M1–M14, Track A: Kinematic Scaffolding & Residual Diffusion)*: Secure ethical approval from the Aalto Research Ethics Committee; convene the Deaf Advisory Board with *Kuurojen Liitto ry*. Implement procedural scaffolding with regularized 2-bone arm IK, seeded with **3D-LEX v1.0** [34] and **ASL-LEX 2.0** [36]. Deploy Continuous Residual Diffusion with localized tolerance bubbles ($r_{\text{tol}} \leq 5$ mm), graph invariance ($\mathcal{L}_{\text{graph}}$), and pain loss ($\mathcal{P}_{\text{pain}}$). Fine-tune on How2Sign [10].
 - *WP2 (M1–M18, Track B: Streaming Perception & Causal Spatial Memory)*: Developed concurrently with Track A. Deploy HaMeR hand tracking and sternum normalization. Build the $\mathcal{O}(1)$ Causal Spatial Memory Bank (M_t) with soft exponential decay, gaze-pointing ray convergence, and θ -role matching resolving regular, backward (INVITE), and orientation verbs (LOOK-AT) via aiming vector $\vec{v}_{\text{aim}}$.
 - **Paper 1 Deliverable (M20): Anatomically Constrained Residual Diffusion for Articulated Sign Kinematics. Target Venues: CVPR / ICCV.**
- **Year 3: Track Convergence & Spatial-SignLLM Foundation Modeling (Core Foundation)**
 - *WP3 (M19–M28, Track B/C: Dual-Rate Streaming & Unified Vocabulary)*: Deploy fast 60 Hz kinematics coupled with slow 1 Hz scene context ($\mathbf{z}_{\text{scene}}$) and entropy-gated active visual

interrupts (v_{patch}) for depicting classifier disambiguation. Unify phonological primitives and 3D coordinate tokens into $\mathcal{V}_{\text{total}}$.

- ▶ **WP4 (M25–M36, Track C: Spatial-SignLLM Pretraining on EuroHPC LUMI):** Pretrain autoregressive Spatial-SignLLM on CSC’s **EuroHPC LUMI supercomputer** (AMD MI250X nodes) across How2Sign and Motion-X++ (19.5M frames) [46] using PyTorch FSDP and ROCm FlashAttention. Pretrain 1–2 Hz macro-compiler and depiction classifier synthesis for OOV terms.
- ▶ **Paper 2 Deliverable (M32):** *Spatial-SignLLM: 3D Spatial-Lexical Foundation Modeling for Continuous Sign Language Translation*. **Target Venues:** ACL / EMNLP.
- ▶ **Year 4: Cross-Lingual Portability, Open-Domain Everyday Chat & Defense**
 - ▶ **WP5 (M37–M42):** *Cross-Lingual Portability & Open-Domain Everyday Chat Benchmark*. Adapt sub-lexical dictionaries from ASL to Finnish Sign Language (**SVK**). Couple Spatial-SignLLM with residual diffusion for closed-loop bidirectional translation. Evaluate on an **Open-Domain Everyday Conversational Benchmark** (evaluating multi-turn dialogue, spatial storytelling, OOV object descriptions, and QA) in collaboration with native Deaf signers from *Kuurojen Liitto ry*.
 - ▶ **Paper 3 Deliverable (M42):** *Multimodal Spatial Foundation Modeling for Open-Domain Assistive Communication*. **Target Venues:** NeurIPS / ICLR.
 - ▶ **WP6 (M43–M48):** *Deaf Community User Study & Dissertation Synthesis*. Conduct formal human-in-the-loop study with certified Deaf/SVK consultants (MOS $\geq 4.2/5.0$). Compile published papers into the doctoral dissertation; public defense at Aalto University.

8 Computational Feasibility, Ethical Governance & Risk Mitigation

8.1 High-Performance Computing Strategy: EuroHPC LUMI Supercomputer

Kinematic pretraining on Motion-X++ (19.5M frames) [46] and Spatial-SignLLM pretraining leverage Aalto’s direct access to the **EuroHPC LUMI consortium** (CSC, Kajaani, Finland):

- ▶ **Hardware & Software Environment:** LUMI-G nodes feature **AMD Instinct MI250X GPUs**. Our codebase is developed natively under **AMD ROCm / HIP**, using ROCm FlashAttention, Composable Kernel libraries, and PyTorch Fully Sharded Data Parallel (FSDP) for multi-node scaling.
- ▶ **Parallel I/O Architecture:** Datasets are hosted on LUMI’s parallel **Lustre filesystem** (/scratch) packaged into indexed **WebDataset** tar shards for high-throughput asynchronous streaming.

8.2 Ethical Governance & Deaf Community Participatory Co-Design

To guarantee ethical integrity and avoid technological paternalism:

- ▶ **Institutional Oversight:** Formal ethical review conducted by the **Aalto University Research Ethics Committee** prior to human participant evaluations.
- ▶ **Participatory Co-Design:** In partnership with *Kuurojen Liitto ry* (Finnish Association of the Deaf), an ongoing **Deaf Advisory Board** provides regular feedback on blendshapes, cultural nuance, and linguistic naturalism from Month 1.

8.3 3-Tier Contingency Fallback Hierarchy

To guarantee timely completion adhering to Aalto’s 3-paper doctoral requirements:

- ▶ **Tier 1 (Core Non-Negotiable):** Full ASL bidirectional procedural-diffusion pipeline, causal memory bank, mathematical loss formulations, and automated quantitative benchmarking (Years 1–3).
- ▶ **Tier 2 (Cross-Lingual Flexibility):** If SVK data extraction encounters delays, cross-lingual transfer focuses on calibrated agreement benchmarks in ASL before scaling to open-domain SVK continuous discourse.
- ▶ **Tier 3 (User Study Scalability):** If user studies face recruitment bottlenecks, evaluation transitions flexibly to a focused panel of 5 certified Deaf interpreters using standardized Likert rubrics.

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